

- 15 In a mixture of two monatomic ideal gaseous X and Y, the molecules of Y have thrice the mass of those of X. The mixture is in thermal equilibrium and the molecules of Y have a mean translational kinetic energy of E_K .

What is the mean translational kinetic energy of the molecules of X?

A $1/3 E_K$

B $1/2 E_K$

C E_K

D $3 E_K$

$$T_X = T_Y$$

$$\langle KE \rangle_X = \langle KE \rangle_Y$$

$$\text{since } T \propto \langle KE \rangle$$

Ans: C

16 The given graph shows the variation with displacement x of the potential energy U of a particle